

Diabetes Prediction Using Machine Learning Classification Algorithms

Assignment Phase 1 & 2: Paper Analysis and Implementation

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1. Introduction & Motivation

Why Predict Diabetes?

- Chronic diabetes is one of the most common diseases worldwide.
- Early diagnosis alongside suitable treatment helps patients restore to a better condition.
- Making the diagnosis process simpler and quicker has a huge effect on the treatment process.

The Role of Machine Learning

- Machine learning techniques are increasingly used to forecast or diagnose life-threatening illnesses.
- Objective: Find the best machine learning classifier with the highest accuracy for diabetes prediction.

2. Dataset Overview

UCI Pima Indians Diabetes Database

- **Total Records:** 768 patient records.
- **Class Distribution:**
 - 268 cases are "tested positive" (Diabetic).
 - 500 cases are "tested negative" (Healthy).

3. Dataset Features

The dataset relies on 8 diagnostic attributes to predict the Outcome:

- ➊ **Pregnancies:** Number of times pregnant.
- ➋ **Glucose:** Plasma glucose concentration.
- ➌ **Blood Pressure:** Diastolic blood pressure (mm Hg).
- ➍ **Skin Thickness:** Triceps skinfold thickness (mm).
- ➎ **Insulin:** 2-Hour serum insulin.
- ➏ **BMI:** Body mass index.
- ➐ **Pedigree:** Diabetes pedigree function (genetic risk).
- ➑ **Age:** Patient's age in years.
- ➒ **Outcome:** 0 (Negative) or 1 (Positive).

4. The Missing Data Problem

Data Preprocessing is Critical

- Reliable outcomes and efficient prediction require clean data.
- There are no standard missing values (NAN) in the dataset.
- **The Issue:** Features like Glucose, Blood Pressure, and BMI have a value of zero in some rows.
- A blood pressure or BMI of zero is medically impossible and meaningless.

5. Preprocessing Strategy

How the Authors Handled Zeros:

- **Step 1:** Identify columns with invalid zero values.
- **Step 2:** Calculate the mean and median for these columns.
- **Step 3 (Crucial):** Perform calculations *separately* for diabetic and non-diabetic patients.
- **Step 4:** Replace the zeros according to the patient's class.

Data Splitting: After preprocessing, data was split into 70% for training/validation and 30% for testing.

6. Classification Algorithms

The paper, and our practical implementation, focuses on these machine learning classifiers:

- **K-Nearest Neighbors (KNN)**
- **Random Forest (RF)**
- **Support Vector Machine (SVM)**
- **Decision Tree (DT)**

7. Algorithm Concepts (Part 1)

K-Nearest Neighbors (KNN)

- Classifies objects based on the nearest samples in the input space.
- It finds the 'k' closest elements in the training set and assigns the most common class.

Random Forest (RF)

- Creates a "forest" made up of many Decision Trees.
- Selects random samples, builds a tree for each, and uses the most votes for the final prediction.

8. Algorithm Concepts (Part 2)

Support Vector Machine (SVM)

- Tries to locate an ideal hyperplane to separate the classes.
- Maximizes the interval between the classes and the hyperplane.

Decision Tree (DT)

- Uses a top-down tree structure to categorize instances.
- Partitions nodes from the topmost root down to the class node.

9. Evaluation Metrics

To evaluate the models, we use a Confusion Matrix (True Positives, True Negatives, False Positives, False Negatives).

The formulas are:

- **Accuracy:** $Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** $Precision = \frac{TP}{TP+FP}$
- **Recall:** $Recall = \frac{TP}{TP+FN}$
- **F1-score:** $F1-score = 2 \times \frac{Recall \times Precision}{Recall + Precision}$

10. Paper's Findings

According to the original study by Nahzat and Yağanoğlu (2021):

- **Random Forest (RF)** was the most successful algorithm.
- It achieved **88.31% Accuracy**.
- It also achieved 88% Precision and 87% F1-Score.
- SVM yielded the lowest performance.

11. My Python Implementation: Setup

Tools Used:

- **Python** via Google Colab.
- **Pandas & NumPy** for data preprocessing and missing value imputation.
- **Scikit-Learn** for modeling and generating classification reports.

I applied the exact class-based median imputation strategy discussed in the paper, maintaining a 70/30 train-test split.

12. My Python Implementation: Results

I successfully reproduced the experiment. My results closely match the original paper:

Algorithm	Accuracy	Precision	Recall	F1-Score
Random Forest (RF)	87.01%	0.86	0.85	0.85
K-Nearest Neighbors	86.15%	0.85	0.84	0.84
Decision Tree (DT)	83.55%	0.82	0.81	0.82
SVM	83.12%	0.81	0.83	0.82

Note: Precision, Recall, and F1-scores represent the macro average from the classification report.

13. Conclusion

- Machine learning has the potential to change diabetes risk prediction.
- Handling meaningless zero values conditionally based on class significantly improves model performance.
- Both the original paper and my Python implementation confirm that **Random Forest** is the most effective algorithm for this dataset.

Thank You!